HEMANTH HARIHARAN
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SCHOLASTIC ACHIEVEMENTS

Program	Academic Institution	GPA	Completion Date
MS, Sustainability	Stanford University	4/4	Jun 24'
B. Tech, Engineering	Indian Institute of Technology Madras	9.17/10	Aug 20'

SKILL SET

Derivatives valuation and trading, Data analytics and visualization, Financial and risk modeling, Monte Carlo Simulations

PROGRAMMING LANGUAGES / TOOLS

Python, SQL, Visual Basic for Applications (VBA), Microsoft Excel

WORK EXPERIENCE (Click links to see details)

Quantitative analyst at NextEra Energy Marketing

Dec 24' – Present

Analyst - Risk Analytics desk on the trade floor at NextEra Energy Marketing

• Valuation of variable-volume OTC swap transactions (notional value up to $1.5 B) primarily hedged using Asian swaps and options.
• Conducted 4 descending-price clocked auctions – analyzing bidding behavior, incorporating dynamic price decrements to optimize margin/headroom under time-dependent pricing.
• Extensive use of data analytics, Monte Carlo simulations (geometric Brownian motion), multi-variate regression (weather normalization), financial modeling etc.
• Collaboration with the futures, options, basis and renewable energy credit trading desks.
• Exposure to Mark-to-market (MTM) accounting practices, portfolio hedging, exotic derivatives (Quanto products, weather derivatives, synthetic financial products), automation bot for Dutch auctions, algorithmic and proprietary trading, price discovery, risk management (position limits, VaR) etc.

Quantitative Research/Model Development

• Fine-tuned derivation of closed-form solution used by team for valuation and hedging Greeks of variable-volume swaps.
• Partnered with oil trading desk head on an alternative formulation for valuation of variable-volume swaps by expressing volume as a function of price. Built proof of concept for dynamic hedging and Greek monitoring.
• Implemented analytical approximations for Asian swap and option payoffs for hedge strategy analyses.
• Developed a simulation tool prototype of mock descending-price auctions.
• Collaborated with transmission traders on fine-tuning model for valuation of exotic congestion derivative (Financial Transmission Rights – FTRs / Auction Revenue Rights - ARRs).

MS, Stanford University

Sep 22' – Jun 24'

Coursework/Research (Project summary and code)

• Deep learning-based modeling of carbon storage
• Machine learning for wind turbine output forecasting
• Convex optimization
• Financial risk modeling of infrastructure project

Internships

COI Energy: Financial modeling for evaluating payback period and forecasting cash flows.
Cypress Creek Renewables: Analyzed trends in publicly available historical interconnection queue data and used machine learning to predict project outcomes.

B. Tech, IIT Madras

Aug 16' – Jul 20'

Coursework/Research

Courses: Probability and Statistics, Differential Equations, Functions of several variables, Series and Matrices, Finite Element Analysis, Fundamentals of Operations Research

Projects/Research:
• Research Internship – GUI development on MATLAB
• Undergraduate thesis – Machine learning to characterize behavior of masonry spandrels.